

The 9th International Scientific Conference
eLearning and software for Education
Bucharest, April 25-26, 2013
10.12753/2066-026X-13-115

**GAME AI ARCHITECTURE FOR TEACHING THE ENGINEERING
DESIGN PROCESS**

Z. KOSMADOUDI, T. LIM, J. M. RITCHIE, R. C. W. SUNG, Y. LIU

*School of Engineering & Physical Sciences, Heriot Watt University, Edinburgh EH14 4AS, UK
zk18@hw.ac.uk, t.lim@hw.ac.uk, j.m.ritchie@hw.ac.uk, r.c.w.sung@hw.ac.uk, ying.liu@hw.ac.uk*

Abstract: *This research paper discusses a research project that employs game's Artificial Intelligence (AI) to improve the ability of engineers to more effectively construct an action plan to reach a solution to a design problem. The engineering design process is modelled through the use of "agents" employing Goal Oriented Action Planning (GOAP), a type of AI that has had success in commercial computer games such as F.E.A.R. The GOAP logic can be described as a cognitive information process, where knowledge is assumed to be predictable and explained with mechanistic information processing.*

Keywords: *Engineering design process, games AI, GOAP, cognition*

I. Introduction

Engineers' main focus is to apply their scientific knowledge to produce solutions to technical problems. Then they can optimize these solutions based on a set of constraints which involve material, technological, economic, environmental and human related requirements. In his paper Penny [1] described engineering design as the "art using science in achieving useful end product" (Fig. 1).

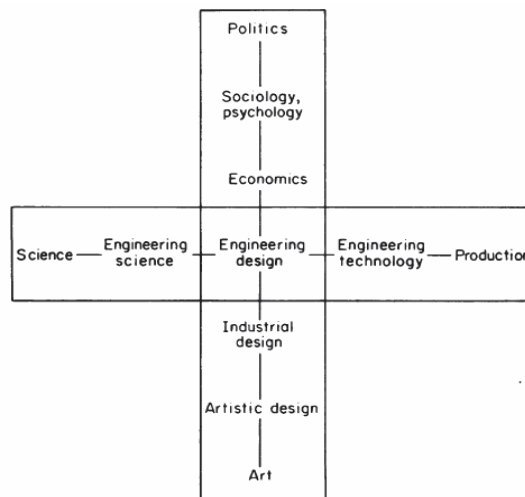


Fig.1 Central Activity engineering design

Engineering design reflects a creative activity that requires a sound knowledge of mathematics, physics, chemistry, mechanics, thermodynamics, hydrodynamics, electrical engineering, production engineering, materials technology, machine elements and design theory [2]. As such there

was been a growing awareness of the importance of engineering design in education. In order to understand and implement effectively the teaching of engineering design, the design processes of engineers have been studied extensively over different approaches [3-7]. Moreover the relationship between engineering design strategy and cognition has attracted much interest; cognition describes the way individuals solve problems, think, perceive, and remember information [8]. However most of the studies generally reach the similar outcome that identifies the design as not an “algorithmic activity”. While engineers develop a design, the goals, problems and constraints are not particularly defined and are highly changeable towards reaching a solution. There might be general guidelines for them to follow in the design process, however there is no consensus regarding one correct procedure or correct strategy to follow in order to reach a solution [9-10].

So far studies pertaining to the design process and cognition in engineering have utilised sketches, drawings, written communication, hands on model building etc., however there less in the literature investigating whether a game theory approach can contribute to a better understanding of the engineering design process, and facilitate in a more effective teaching of design courses. This research paper was designed to investigate whether a game AI module of goal oriented action planning (GOAP) which was first implemented in the commercial game called F.E.A.R, for developing its characters behaviour, can align with the engineering design process.

1. Background

In engineering design, problem representation is highly significant [11-12], since design problems are complex, have many solutions and often involve unidentified elements, incorporate multiple domains and have non-engineering constraints [13]. Developing a creative solution is often a matter of restructuring the problem space by reorganizing existing knowledge or transforming concepts [9-10, 14-15].

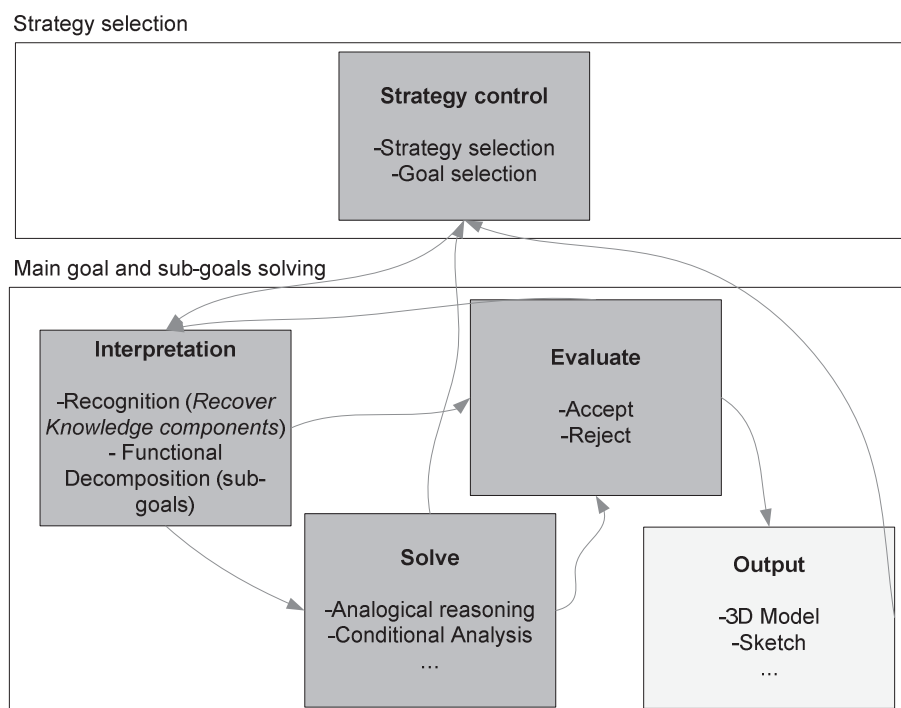


Fig. 2 Heuristic representation of information processing stages

Problem decomposition and solution composition are fundamental in design processes [2]. Goel et al.[16], in his experiments observed designers identify a problem and use a number of strategies (can be a reference to behaviours as an action concept) to reframe main functions (goals, requirements of the problem) to more abstract functional terms (sub-goals) with pre-conditions

(interdependencies) and its post-conditions (effects). The authors also observed that designer's new concepts were generated by composing multiple cross platform analogies and by using a combination of textual descriptions (pictures, graphs, and mathematical representations) (Fig. 2).

The current approach in problem decomposition is focused on the design cognition [17]. In cognitive science, decomposition is described as a technique of breaking problems into more transparent and less-complex parts that can be solved separately. These parts are called sub-problems or sub-goals which usually enclose different kinds of interdependencies [18] and can be represented as hierarchies of function structures [19]. Knowledge is mandatory for problem decomposition, because in order to divide a problem, something must be known about it. As Lloyd and Scott [20] suggested;

“[the] specific experience of the problem type enables designers to perceive new problems through old solutions, and to structure and decompose a design problem”.

1.1 Learning Theories

The theory of problem solving or thinking or learning is a theory of the environment of task [21], where there are stimuli, possible actions, and goals which drive a behaviour (also described in terms of “knowledge components”) [21].

The experiential learning theory (ELT) describes a process where knowledge is created by transforming experiences [22]. In their research Kolb and Fry [22], and later Kolb [23] present the ELT as a model consisting of concrete experience, observation and reflection, abstract concepts and active experimentation. Their general conclusion is that an individual learns through reflecting on effects on a particular action, in a particular setting, and though attempting to understand those effects and modify that action to accommodate a new idea. Kolb [23] also mentioned that “learning is a process whereby knowledge is created through the transformation of experience” (Fig. 3).

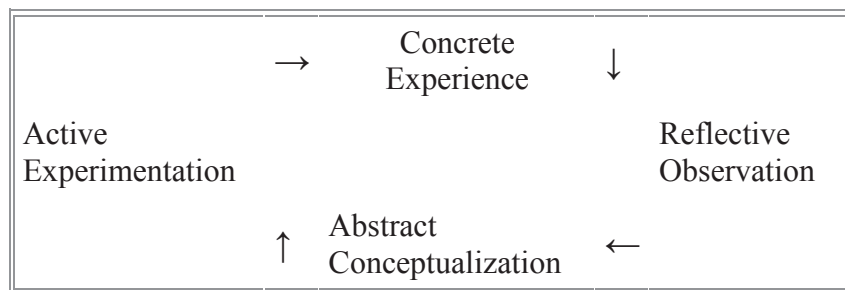


Fig.3 Experiential learning theory (ELT) process

Similar to experiential learning theory, the constructivism approach to learning adopts methodologies consistent with the principles of relativistic interpretation. An individual's knowledge of reality is not accurate and objective; his knowledge is subject to his experience, knowledge and cognitive processes. Knowledge and learning is constructed through the cognitive interaction of individuals with their social environment, and modified by the inclusion of new data [24].

The above theories suggest that not only “experiences” have an effect in learning, but also the testing of ideas and receiving feedback (performing actions and evaluating results, cognitive interaction) to alter the ideas and construct new knowledge.

Experiential learning and constructivism theories provide a good base for integration of gameplay. Phillips [25] focused on constructivism and behavioural learning and defined learning as a construction of cognitive structure through actions on a game world. Gredler [26] categorized simulations and learning games for experiential training. Amory and Seagram [27] proposed the integration of educational theory and game design aspects. Kivinen&Ristelä[28] explored the constructivist view of acquiring knowledge on terms of actions; the authors considered knowledge as description nodes which are being replaced only when, during their evaluation, there are other more

useful descriptions for that particular purpose. This suggests learning can be defined as a construction of cognitive structures through action in a game world.

1.2 GOAP

The game AI module of Goal oriented action planning (GOAP) aligns well with the learning theories in 1.1.

GOAP is an AI planning architecture which has been designed and implemented for games such as F.E.A.R. GOAP is based on STRIPS (STanford Research Institute Problem Solver) which is an automated planner including initial state and goal stages. GOAP is a decision-making planner; searches agent's next actions. Specifically, GOAP is defined in terms of goal, action, plans and formulate. There can be a number of goals that an agent wants to achieve. Once one goal is active, determines the characters / users behaviour. A goal's structure determines its current relevance, and knows whether or not has been satisfied. A plan is sequences of actions, and can be identified as a valid one when it satisfies a goal and take the agent from some starting state to some state that satisfies the goal. An action is a single, atomic step within a plan that makes an agent "do" something. Each action has preconditions and effects. Preconditions and effects provide a mechanism for chaining actions into a valid sequence. In GOAP, each action has a static cost value calculated from the cost function, through which the planner can determine the next action. A goal-directed character/user displays some measure of intelligence by autonomously deciding to activate the behaviour that will satisfy the most relevant goal at any instance. During the gaming procedure, a character that formulates its own plan to satisfy its goals exhibits less repetitive, predictable behaviour, and can adapt its actions to custom fit the current situation. In addition, the structured nature of GOAP architecture facilitates authoring, maintaining, and re-using behaviours [29].

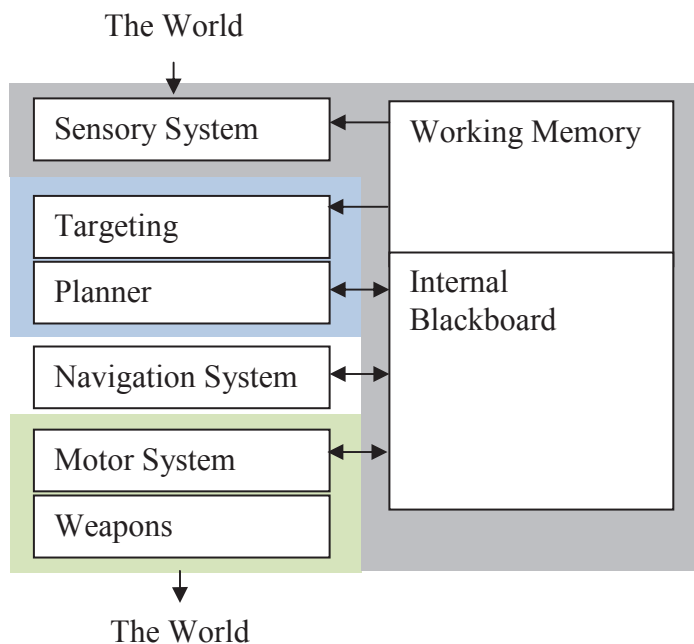


Fig. 4 F.E.A.R. agent brain architecture, arranged into three sections: a “perception and memory”, “action selection”, and “motor and navigation”

In general GOAP is an AI architecture that was implemented to facilitate the demand of the gaming community for games that incorporate more intelligent and engaging characters, which are more robust, reactive, adaptable, expressive, sensible and scalable [30]. As such, GOAP unifies theories of cognition; in the game F.E.A.R. - , in particular, the GOAP agent architecture (Fig. 4) employed a modified version of the cognitive architecture as described by the Synthetic Characters Group or C4 from MIT [30].

GOAP's architecture allows a character to decide not only what to do, but how to do it. GOAP can both constrain the actions of an engineer as well as afford the construction of useful action plans. This architecture also enables an automated strategy checking. In addition, it enables the construction

of strategies in form of “agents” that guide the engineer in constructing an action plan as well as critique the constructed plan. GOAP, in this way, can be described as a cognitive information process, where knowledge is assumed to be predictable and explained with mechanistic information processing (respond to input of data) [31].

II. An illustrative example of GOAP in an engineering task

To describe how the static semantics of a cognitive task analysis enables interactive construction of GOAP models for engineering design process, let’s us consider a simple example: peg in the hole (Fig.5).

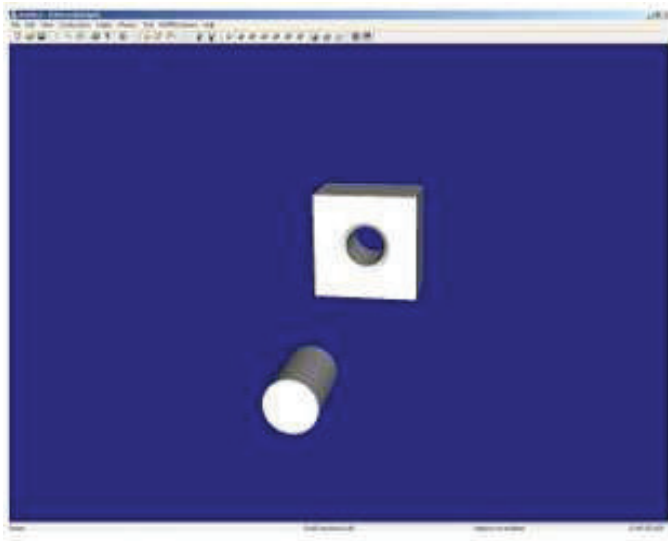


Fig. 5 Peg in the hole

2.1 Strategies, Behaviour & Functions

GOAP can be approached as a designer’s behaviour strategies (agents) composed of entities of pre-existed knowledge which can be associated with context of a set of functions and their pre-conditions. A function can have many pre-conditions associated with it. Each condition may describe a function’s affect. Once a strategy is active in determines the user behaviour which in return contains a) states which are related to a function’s pre-conditions, b) transitions, which are relationships between two states. The states in behaviour specify the evolution in the values of the conditions of a function. Each state transition in behaviour is annotated by the causes for the transition. Causal explanations for state transitions may include, for example, conditional constraints.

In the example of a peg in a hole there is mainly a single strategy of “peg in the hole”, that compromises of the main goal of “insert”. However to reach that goal, sub-goals need to be composed, such as peg alignment with the hole, contact and place peg in the hole and peg’s size to fit in the hole diameter. Reaching the end goal of inserting the peg in the hole, the “peg” and the selected strategy needs to go through different states defined by the preconditions of the associated functions; fit, place, contact, and alignment. The transition between the states is based on the conditional statements (Fig. 6).

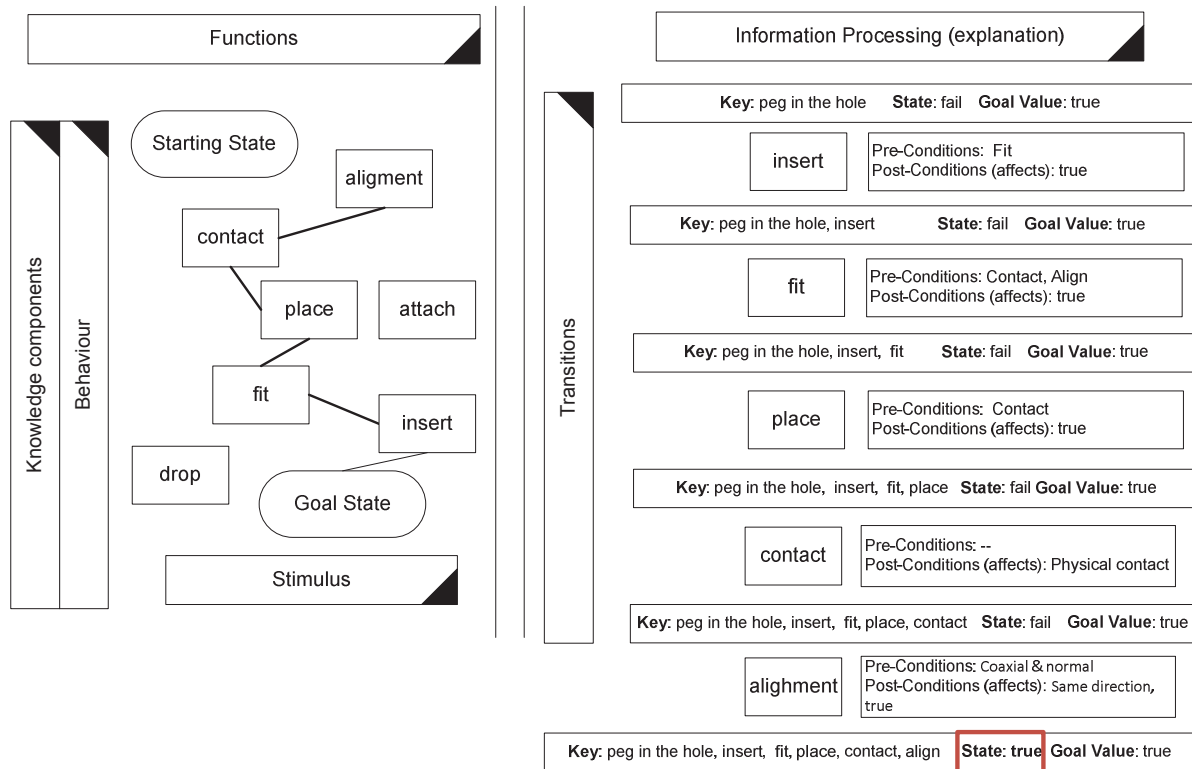


Fig.6 “Peg in the hole” planning problem

III. Conclusions

GOAP as a game AI architecture envisioned to provide some level of support in teaching engineering design process. GOAP enables designers to decompose complex engineering problems to the levels of abstractions provided by the “perception and memory”, “action selection”, and “motor and navigation and thus provide the potential for learning different things at different levels of abstractions of the problem. Fundamentally, dividing learning problems between systems makes learning easier.

In fact, a major point of GOAP inspired engineering design process is to think out of the box of the well-trodden problem definitions in engineering, and explore new ways of thinking about problems. Unlike conventional methods such as brainstorming, browsing the Internet, or checking engineering textbooks, GOAP can provide means to rapidly access systems functionally and at multiple levels of abstraction. The marriage of strategies analogy and functional indexing enables the development of creative design solutions to difficult problems. Finally GOAP as an AI architecture is grounded in cognitive studies.

There are a large number of cognitive studies in games and educational sciences as well as very large number of studies of engineering design processes. The authors believe the time has come to bridge the two lines of research, by grounding the development and assessment of engineering design through serious games and cognition studies. For future research directions are to demonstrate GOAP case studies with more complex decision making processes during engineering sessions in a game environment.

References

- [1] Penny, R.K., (1970). Principles of Engineering Design. Postgraduate 46, pp. 344–349.
- [2] Pahl G., Beitz W., Feldhusen J., Grote K.-H., (2007). Engineering Design A Systematic Approach 3rd Edition.
- [3] Atman, C., Cardella, M., Turns, J., Adams, R., (2005). Comparing freshman and senior engineering design processes: An in-depth follow up study. Design Studies, 26(4), pp. 325-357.
- [4] Christiaans, H., Dorst, K., (1992). Cognitive models in industrial design engineering: a protocol study. Design Theory and Methodology, 42, pp. 131-137.
- [5] Kruger, C., Cross, N., (2006). Solution driven versus problem driven design: strategies and outcomes. Design Studies, 27(5), pp. 527-548.
- [6] Ennis, C., Gyeszly, S., 1991. Protocol analysis of the engineering systems design process. Research in Engineering Design, 3(1), 15-22.
- [7] Radcliffe, D., Lee, T., (1989). Design methods used by undergraduate engineering Students. Design Studies, 10(4), pp. 199-207.
- [8] Cassidy, S., 2004. Learning Styles: An overview of theories, models and measures. Educational Psychology, 24(4), pp. 419-444.
- [9] Cross N., (2001). Design cognition: Results from protocol and other empirical studies of design activity C. Eastman, M. McCracken, W. Newstetter (Eds.), Design knowing and learning, Elsevier, New York, Pages 79–103.
- [10] Zimring, C., Craig, D., (2001). Defining design between two domains: Argument for design research á la carte. In C. Eastman, M. McCracken, & W. Newstetter (Eds.) Design Knowing and Learning Elsevier: New York, NY.
- [11] Cross, N., Cross A., 1998. Expertise in engineering design. Research in Engineering Design, 10, Pages 141-149.
- [12] Mehalik, M., Shunn, C., (2006). What constitutes good design? A review of empirical studies of design processes. International Journal of Engineering Education, 22, Pages 519- 532.
- [13] Jonassen, D., Strobel, J., Lee, C., (2006). Everyday problem solving in engineering: Lessons for engineering educators, Journal of Engineering Education, 95(2), Pages. 139-151.
- [14] Ansborg, P., Dominowski, R., (2000). Promoting insightful problem solving. Journal of Creative Behaviour, 34, pp. 31-60.
- [15] Mumford, M., Baughman, W., Sager, C., (2003). Picking the right material: Cognitive processing skills and their role in creative thought. In M. Runco (Ed.) critical creative processes, Cresskill, NJ: Hampton Press, pp. 19-68.
- [16] Goel A. K., Vattam S., Wiltgen B., Helms M., (2012). Cognitive, collaborative, conceptual and creative — Four characteristics of the next generation of knowledge-based CAD systems: A study in biologically inspired design, Computer-Aided Design, 44(10), pp. 879-900.
- [17] Eastman, C., (2001). New directions in design cognition: studies of representation and recall in C Eastman, W Newstetter and M McCracken (Eds.) Design knowing and learning: cognition in design education Elsevier, Oxford, UK.
- [18] Simon, H. A., (1996). The sciences of the artificial, 3rd Edition, MIT Press, Cambridge, M. A.
- [19] Smith, G. F., Browne, G. J., (1993). Conceptual foundations of design problem solving, IEEE Transactions on Systems, Man and Cybernetics, 23, Pages 1209-1219.
- [20] Lloyd, P., Scott, P., (1994). Discovering the design problem, Design Studies, 15 pp. 125-140.
- [21] Newell, A., Simon, H. A., (1972). Human problem solving. Englewood Cliffs, NJ: Prentice-Hall.
- [22] Kolb D. 1984. Experiential learning: Experience as the source of learning and development Prentice-Hall, Englewood Cliffs, NJ.
- [23] Kolb D., Fry R. Toward an applied theory of experiential learning C. Cooper (Ed.), Theory of group process, John Wiley, London, UK.
- [24] Raptis, A., Raptis, A., (2004). Learning and Teaching in the Information Society: Educational Activities, A&B, Athens.
- [25] Phillips, D. C., (1995). The good, the bad, and the ugly: the many faces of constructivism. Educational Researcher, 24, pp. 5–12.
- [26] Gredler, M. E., (1996). Educational games and simulations: a technology in search of a (research) paradigm. In D. H. Jonassen (Ed.), Handbook of research for educational communications and technology. New York Simon & Schuster Macmillan.
- [27] Amory, A., Seagram, R., (2003). Educational game models: Conceptualization and evaluation. South African Journal of Higher Education, 17, pp. 206–217.
- [28] Kivinen, O., Ristela, P., (2003). From constructivism to a pragmatist conception of learning. Oxford Review of Education, 29, pp. 363–375.
- [29] Orkin J., (2006). Three states and a plan: The AI of F.E.A.R. In Proceedings of the Game Developer's Conference (GDC).
- [30] Burke, R., Isla D., Downie M., Ivanov Y., Blumberg B., (2001). Creature smarts: The art and architecture of a virtual brain. In the proceedings of the Game Developers Conference. San Jose, CA, Pages 147-166.
- [31] Newel A., (1990). Unified theories of cognition, Harvard University Press.

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.